

[From the Field]

Carbon Profits or Pastoralist Precarity: The 'Sale of Air' in Northern Kenya's Double-Edged Climate Financing Frontier

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Abstract : The global push to mitigate climate change has propelled carbon credits as a critical market tool for offsetting emissions. However, the Northern Kenya Rangeland Carbon Project (NKRCP) highlights a “double tragedy” whereby the apparent benefits of carbon revenues often conceal deeper issues of social disparities. Despite the designed aim to improve rangeland management and generate revenue, NKRCP restricts pastoral access to traditional land, exacerbating violent conflicts over natural resources in an area vulnerable to climate change. Drawing from qualitative data collection approaches in Samburu and Isiolo Counties, the findings show that the economic benefits from carbon credits disproportionately benefit brokers and local elites rather than the pastoral communities directly affected by the intensification of land use change. The empirical contribution of this paper sheds light on the increasing complexities surrounding market-driven nature-based solutions in expansive and resource-rich marginalized regions. These regions continue to be subjected to the exploitation and extraction of natural resources for capitalist interests by the Global North without considering the well-being of Indigenous populations in the climate-sensitive areas.

carbon credit

pastoral communities

northern Kenya

conflicts

Northern Rangeland Trust

Introduction

Resource conflicts in East Africa have often been rooted in the lifestyle of pastoralist livelihoods that rely heavily on access to water and grazing lands (Mkutú, 2008). In recent years, these pastoralist livelihood systems have been increasingly at risk due to climate change vulnerability and shifts in nature-based economies, such as the emergence of carbon markets through the expansion of conservation initiatives (Holmes, 2018; Evite and Zara, 2023). The United Nations Framework Convention on Climate Change (UNFCCC) argues that climate finance is one of the tools to adapt to the adverse effects of climate change. Article 9 of the Paris Agreement states that international climate finance can be understood as the “provision and mobilization of resources by developed countries to developing countries” (UNFCCC, 2022). Carbon markets are driven by mechanisms such as climate finance, which refers to the money flow to support mitigation and adaptation to climate change (Lugusa, 2020). When financial resources are scarcely transferred to developed countries, it brings the challenge of increasing pre-existing conflicts in vulnerable regions over livelihoods, resources and land rights (Bersaglio, 2017).

This study discusses the case of the North Kenya Rangeland Carbon Project (NKRCP) that illustrates how “conflict and climate change interact by increasing indirect social vulnerability through disproportionate loss and damage by those exposed to both harms” (Buhaug and von Uexkull, 2021). International non-governmental organizations like The Nature Conservancy and Conservation International indirectly control access to and use of natural resources by funding conservation landscapes within Indigenous territories (Mbaria and Ogada, 2017). While these organizations bring economic empowerment and scientific expertise in managing Indigenous resources vulnerable to climate change, their motives are often seen as expansionism of capitalist interests through the privatization and commodification of nature, particularly in the Global South. The perception that conservation is, thus, a “white-man-driven initiative” further exacerbates community disillusionment as unintended consequences.

Northern Kenya Rangeland Carbon Project

The NKRCP encompasses 1,993,075 hectares (4.7 million acres) of grassland in northern Kenya (Pinette, 2024). It is the world’s largest soil carbon removal project managed by the NRT, a non-governmental organization that initiated conservation work in the 1990s. They advocate for community-based conservancies (CBCs), facilitate their management, and promote planned grazing and the conservation of wildlife corridors and forests in Kenya and Uganda. NRT has 44 community conservancies in Kenya, most of which are in northern Kenya, where inhabitants are nomadic pastoralists who share the environment with wildlife. The project that began in 2012 was approved in 2021 by the Verified Carbon Standard (VCS), a program that credits greenhouse gas (GHS) projects and is run by Vera. This non-profit institution manages and develops standards for climate action, responsible business practices, sustainable development and the Climate, Community & Biodiversity Alliance (CCBA) standards. The benefiting CBCs are situated in Laikipia County, Marsabit County, Samburu County, and Isiolo County (see Figure 1). ¹



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The empirical data presented here is part of 75 in-depth interviews and focus group discussions with stakeholders in Samburu and Isiolo County based on 11 months of fieldwork between 2022 and 2024 within 8 of the 14 carbon credit sites under the NRT governance in northern Kenya.

The NKRCP operation entails a rotational grazing pattern different from pastoral communities' traditional animal husbandry. The project monitors the net methane (CH₄) from traditional pastoral livestock activities. It implements a rotational grazing pattern, assisting in storing carbon in the soil and recovering perennial grassland. The project will be implemented for 30 years from 2021 and is currently estimated to positively impact 175,000 nomadic pastoral lives (NRT, n.d.). It is assumed that it provides a novel economic intervention that can reduce environmental pressures by incentivizing sustainable land management (NRT, 2022).

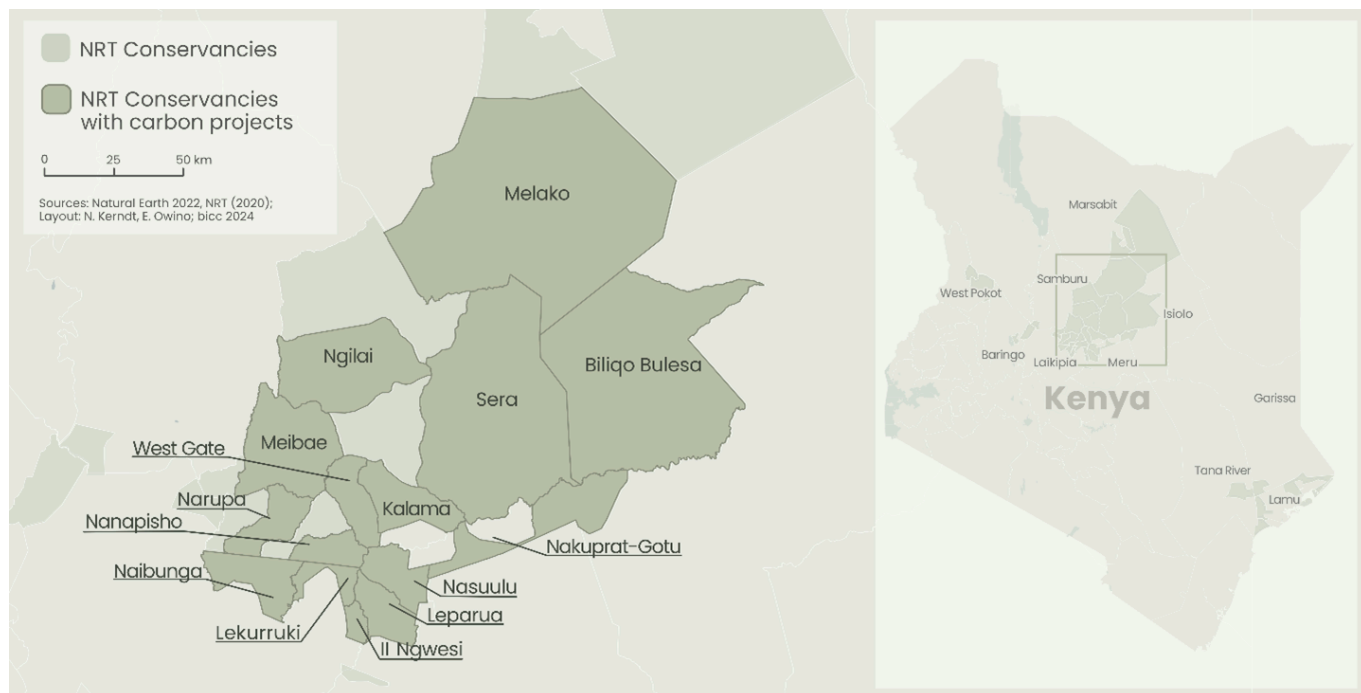


Figure 1: Map of NRT Carbon project area within Laikipia, Samburu, Isiolo and Marsabit Counties in Northern Kenya. modified from (Owino et al., forthcoming)

Double Exposure Framework

Using the double exposure lens, the main question I probe is: “To what extent are the losers in climate change also the losers when it comes to economic globalisation?” (O’Brien and Leichenko, 2000). Political ecologists developed this approach to understand the simultaneous effects of climate change and globalization on regions, communities, and ecosystems. It exposes carbon credits as a double-edged sword, where excessive rain and droughts brought on by climate change intersect with a changing financial landscape in the globalized economy through trade liberalization and increased foreign investments.

Vulnerable communities often lack the capacity to cope with this kind of double pressure, and this usually leads to a worsening of their standard of living and increased economic vulnerabilities unless mitigated by targeted interventions (Evite and Zara, 2023). While carbon credits provide financial incentivization to local communities for maintaining rangelands, swamps or forests to store more carbon and help mitigate climate change, they have proven to pose potential risks of exacerbating resource-based conflicts due to abrupt and large-scale land-use changes and restriction policies that socially and economically affect local communities in marginalized regions. Thus, critics of climate finance argue that it is a mechanism that promotes neocolonialism in the Global South because it is an initiative funded by the Global North (Evite and Zara, 2023). The argument is that the commodification of natural resources, often through carbon credit schemes, has led to unprecedented land use and ownership shifts. This has exacerbated resource-based conflicts in these regions as the authority over land has gone into the hands of private businesses at the frontier and out of the hands of the state (Bersaglio, 2017; Lohmann 2008). These disruptions affect traditional livelihoods and create dependency on profit-oriented external markets from carbon credits.

Northern Kenya's Historical and Current Conflict Dynamics

Today's northern Kenya, formerly known as the Northern Frontier District (NFD), was established during British colonial rule in 1925. It covers a vast area inhabited primarily by pastoralist communities. Throughout the colonial period, the NFD was administered separately from the rest of the country, a policy driven by its perceived remoteness and security

concerns related to the region's proximity to Ethiopia and Somalia (Schlee, 2013). Following a failed attempt to join Somalia after Kenya's independence in 1963, the NFD has coped with a legacy of neglect, marginalization, and contested identity to date (Sharamo, 2014).

Northern Kenya has long been characterized by violent conflicts, particularly among pastoral groups. Using the ACLED (Armed Conflict Location and Event Data) to analyze pastoral conflicts in the last five years, Figure 2 illustrates the fluctuation in lethal events—at least one person killed, the number of conflict events involving pastoral actors and the resulting fatalities. Conflict incidents intensified, and lethal conflicts also showed a substantial increase. This is partly tied to the prolonged drought that affected northern Kenya between 2018 to 2023, suggesting a correlation between environmental-induced challenges that enhanced the mobility of pastoralists and intensifying conflicts over scarce resources. The 2007 conflict also shows the post-election violence that affected the whole country. Traditional ethnic rivalries, the need to restock livestock after drought or disease, and cultural practices such as revenge, supremacy battles, and bride price payments continue to play a role (Witsenburg and Adano, 2014). More recent factors include boundary-making conflicts caused by ethnically defined counties since 2013.

Devolution is pushing for defined county boundaries, escalating ethnic mobilization and politicization of ethnicity. These new dynamics, combined with the commercialization of livestock rustling, have intensified resource competition as fighting is focused on access and control, and boundaries seem to be a tool that can enable access or limit one community against accessing resources across inter-county borderlands (Eaton, 2010; Schilling et al. 2012). Critical pastoral resources such as livestock, water, land, and pasture are central to these conflicts. These resources are all affected by environmental pressures, including droughts, floods, and increasing variability in the region's two major rainy seasons, April and October (Eberle et al., 2019).

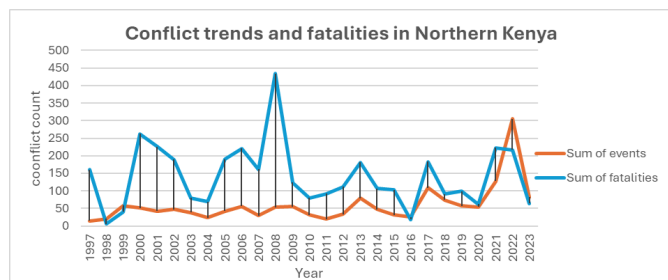


Figure 2: Conflict events and fatalities in northern Kenya between 2018 and 2023 (Source, ACLED, 2024).

In Northern Kenya, while climate-related resource scarcity, such as the prolonged drought, affects the timing and frequency of raids, governance of these resources in the newly devolved system, are critical in determining whether resource depletion leads to conflict or more cooperation. With NRT through conservation initiatives, conflicts intensified following conservancies' expansion and commercialized raids ¹ it is involving local warriors and NRT rangers. The NGO has been condemned by local communities as an agent of violence following the extrajudicial killings of local pastoralists, even though the NRT promotes itself as an agent of peace by defining its work in conservation as peacebuilding (Counsell, 2023).

The Unintended Consequences of NKRCF in their Attempt to “Sell Pastoral Air”

The Natural Solution Climate Alliance (NSCA) and President William Ruto, during the COP27 climate conference in 2022 ², presented NKRCF as a crucial and timely initiative to combat climate change through carbon sequestration while promoting biodiversity and sustainable rangeland use for wildlife and pastoralists. However, beneath the environmental solution agenda lies a problematic complex of social inequalities and violent contestations stemming from restricted grazing policies and the exclusion of pastoralists from carbon financial benefits ³. The reallocation of traditional communal grazing lands for carbon sequestration zones within Borana and Samburu land turned into a violent invasion in 2023 when young morans (unmarried warriors) ⁴, armed with rifles, felt they had no choice but to forcefully access core conservation areas preserved for wildlife and tourist lodges to access pasture for their livestock. Within the carbon methodology, the idea that pastoralists historically depended on the ability to move across vast landscapes in search of water and pasture for their livestock has been completely ig-

nored. Mobility is essential for their survival as resources were shared in times of severe environmental scarcity, which has been disrupted by the carbon credit model of preserving rangelands for carbon storage in the soil (Mkutu, 2008).

Some interviewees expressed frustrations following the restriction policies that exacerbated existing vulnerabilities. Especially during the prolonged drought from 2018 to 2023, which swept almost 80% of pastoral livestock in northern Kenya, the need for pasture was urgent (see the spike of conflicts from Figure 2 above). The exclusion from these grazing lands was perceived as a loss of economic livelihood and a violation of customary land rights ⁵. The situation worsened insecurity and often pushed the state to deploy police officers to protect tourist facilities from local invasions ⁶. Between 2022, when carbon funds were dispatched to conservancies, and 2023, NRT spent more resources and time on peacebuilding activities following a spike of violence in conservancies rather than using those funds for pastoral development. These areas are now converted to tourism sites with expensive lodges that locals cannot access or afford, even for manual labor. They are also defined as core conservation areas for wildlife, blocking the pastoral communities. This led to tensions among the locals who felt disenfranchised by the carbon project, placing international markets over the well-being of the locals and the survival of their culture and livelihood for future generations ⁷. As one of the elders representing the Borana community in Isiolo stated ⁸;

...we have a traditional system of managing rangeland known as the dedha system,...NRT demonized this as an “unplanned” way of managing rangeland; hence, they brought a “planned” rotational grazing that only allows us to graze in one place for a maximum of three weeks...This has undermined our Indigenous knowledge that we fear is now viewed as primitive, and the hidden agenda here is to deny us access to land and grab it...this carbon agenda is a scam of the decade...they are using our name to get money, and nothing comes to the community...

Moreover, inequalities have been exacerbated by the way carbon revenues are distributed. As NRT and international partners like The Nature Conservancy benefit from carbon sales, the local pastoralists reap limited or no financial rewards. For instance, NRT

carbon sales (Ndirangu, 2024). According to the NRT bylaws, the funds should be directed through the conservancy managers, who work with the board members to make decisions for the community regarding which projects or issues to focus resources on and support the conservancy members. However, the locals interviewed felt sidelined from the decision-making process, resulting in increased mistrust, frustration and suspicion about the true beneficiaries of carbon funds. The pastoralists interviewed claimed that the process is top-down and institutionalized, fearing there is no transparency in what NRT gets from the “sale of their air” ⁹.

Interviewees also pointed to systemic issues within conservancies, as unequal distribution of carbon revenues reinforces existing social hierarchies because elites closely supported by politicians and economic power structures benefit disproportionately from carbon credit sales. It was noted that some board members and managers are accused of mismanaging funds, and there are reports of political interference and manipulation during elections for conservancy chairpersons making it a lucrative position to die for ¹⁰. This is because elites often control resources channeled to manage development within conservancies. They allegedly use their positions, power, and influence for personal gain, excluding the broader community from the benefits. The sidelining of pastoralist voices fuels resentment and perpetuates further marginalization in a region grappling with historical neglect policies stemming from colonial and post-colonial regimes that have not been addressed to date ¹¹. Ethnic rivalries, governance challenges, and limited access to development further complicate these resource-based conflicts ¹².

According to Survival International (Blood Carbon report) of 2023 ¹³, the NKRCF project design should have considered the Indigenous knowledge systems and priorities of pastoralists who own land and resources. Locals also affirmed this during the interviews. Some respondents felt that the promise of long-term environmental sustainability through carbon credits was too abstract for local needs and realities. These specifically stemmed from grazing committees and councils of elders unaware of where the carbon revenues disappeared after being released by NRT to the conservancy managers ¹⁴. Many stated that they are left in the dark as no one has ever called for a public participation meeting per the Kenyan constitution to decide on projects appropriate for

pastoralists, as few elites decide on their behalf what they think is best for the community. Another significant issue concerning the concept of the carbon project is the gap in understanding about what carbon is and how it is measured, how it operates, and where it is sold. Pastoralists interviewed could not fully grasp the concept of carbon trading. To them, they know NRT is “selling their air” or “Suwo” (as the Samburu call it) ¹⁵. This has led to confusion, rumors, and mistrust as some were skeptical of taking the money, alluding to it as dirty/blood money because selling air is like selling ghosts ¹⁶.

Some communities also acknowledged receiving benefits such as scholarships and bursaries for their children, water projects, and employment of rangers in conservancies. However, the general sentiment is that these benefits are unevenly distributed throughout northern Kenya, which is conserving its rangelands. The issue of corruption and emerging carbon millionaires who are conservancy managers is becoming a concern ¹⁷. The perception is that a significant amount of money is being funneled into private hands rather than shared equitably by the communities doing the rangeland management work ¹⁸. When carbon was implemented, the pastoral communities had higher expectations and aspirations, hoping to improve their livelihoods. Although the project was initially perceived as a potential source of wealth and development, the reality has disappointed many pastoralists ¹⁹.

Conclusion

The surge of capitalist nature-based solution projects such as carbon credits poses a risk of escalating natural resource conflicts, especially in the Global South, where rangelands, forests and swamps are targeted and converted in the name of invasive and expansive activities such as carbon offset schemes (Weldemichel, 2022). The findings reveal that climate finance initiatives, such as carbon markets, offer new economic opportunities to empower Indigenous communities' livelihoods. They have proven that they also limit control, access, and use of critical resources like rangelands for pastoralist groups in northern Kenya. This complicates traditional resource-sharing mechanisms by the pastoral communities, whose ethnic or county boundaries have never been delimited. The carbon project poses the risk of

dispossession of pastoralists from their ancestral lands, amplifying resource competition by creating scarcity in fragile ecologies prone to climate variabilities. Several communities had to alter traditional grazing patterns due to climate variability and newly imposed land management policies under the carbon sequestration project promoted by NRT. For communities that have faced marginalization and underdevelopment for decades from colonial to post-colonial regimes in Kenya, the carbon project is seen as another form of invasion and alienation of Indigenous land.

External entrepreneurs impose carbon credit interventions that do not serve pastoralists' interests nor address their urgent realities and needs. The entanglement of harsh environmental changes, natural resource exploitation and restrictions, and social inequalities intensifies competition over shrinking resources. These communities are already locked in ethnic rivalries and resource-based conflicts. They are now experiencing the new dilemma of greater polarization as restrictions on land are viewed through the lens of carbon colonialism. Additionally, the intensification and expansion of conservation efforts in northern Kenya continue to pose further challenges produced by international actors' capitalist-driven motives, the lack of genuine community involvement, the missing action of the Kenya government in protecting the vulnerable communities, the inadequate devolution of conservation landscapes, and ongoing resource conflicts. These issues undermine carbon markets' professed sustainable environment, wildlife conservation, and equitable and sustainable development goals. The findings have shown that the NKRCP exemplifies a double exposure tragedy in that pastoral communities grapple with the double-edged frontier of climate-induced environmental changes and the commodification of capitalist appropriation of natural resources.

1 Commercialised raids are economically driven, organized cattle-raiding activities that have evolved from traditional subsistence practices into profit-oriented operations, driven by factors such as market demand, availability of firearms, involvement of cartels, and the influence of political actors leveraging territorial competition and resource control (Greiner, 2013; Mkutu, 2008)

2 See <https://www.nrt-kenya.org/news-2/2022/12/16/the-northern-kenya-rangelands-carbon-project-a-community-based-climate-solution-10-years-in-the-making>

3 Interview with a grazing official at Westgate Conservancy March 13, 2023.

4 In East African cultures within Maasai and Samburu tribes, morans refers to unmarried warriors who have undergone rites of passage and responsibilities of guarding communal wealth-livestock and defending the community against external attacks by enemies (Mkutu, 2008).

5 Focus group with herders in Kalama Conservancy March 6, 2024.

6 Interview with senior police commander in Wamba town February 6, 2024.

7 Ibid

8 Interview with Borana council of elder at Isiolo town April 17, 2024.

9 Focus group with Borana women April 28, 2024.

10 Focus group discussion with elders in Gotu April 29, 2024.

11 Interview with civil society of Kenya official in Nairobi May 21, 2024.

12 Ibid.

13 See Simon Counsell's report for Survival International from 2023, Blood Carbon: how a carbon offset scheme makes millions from Indigenous land in Northern Kenya.

https://assets.survivalinternational.org/documents/2466/Blood_Carbon_Report.pdf

14 Focus group discussion with Borana women April 28, 2024.

15 Interview with a local activist in Maralal town January 29, 2024.

16 Focus group discussion with community members in Namunyak conservancy-Ngilai Central March 6, 2024.

17 Ibid.

18 Interview with women group leader at Kalama conservancy working on planned grazing program May 17, 2024.

19 Focus group discussion with Borana women April 28, 2024.

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